BOMBARDILO CROCODILO

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PROBLEM STATEMENT

Hill-Station Logistics Drone

Mission: Replace porters and mules on steep terrain by carrying 10 kg to an altitude of 3,000 m.

Requirements: 10 kg payload; 8 km one way; account for the thrust and endurance loss at altitude; wind 12 m/s with ridge turbulence; terrain-following flight; takeoff weight under 25 kg.

Justify: propulsion sizing for density altitude, energy source, and the safe delivery method on slopes.

Deliverables: design report (5–8 pages) with sizing calculations, CAD assembly, analysis/simulation results, weight–power–endurance budget, cost estimate, 5-minute pitch.

GENERAL SOLUTION



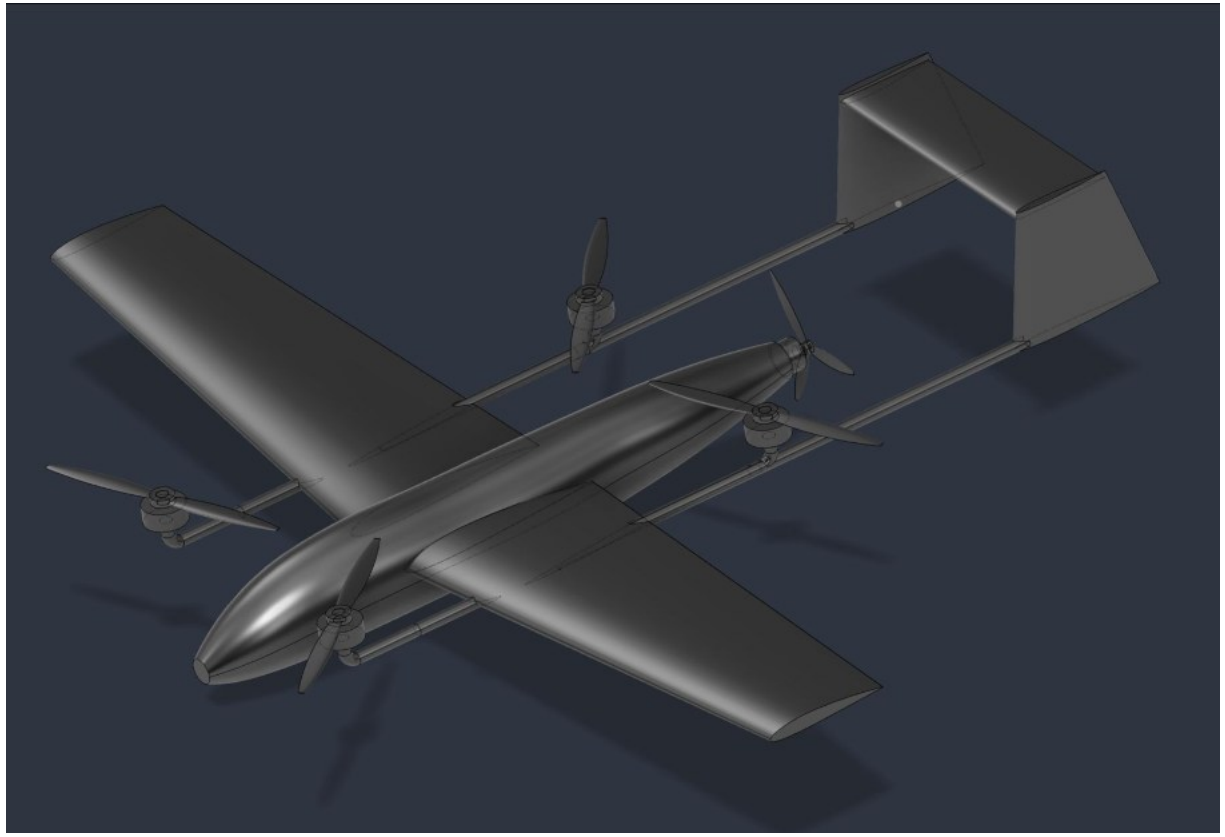
Multirotor/Quadcopter



Fixed Wing UAV

QUADPLANE VTOL





GENERAL PHYSICS

- Increasing elevation = Reduced air density, $\rho_{3000} \rightarrow 25.8\%$ reduction with respect to ρ_0
- Leads to loss of thrust, hence thrust to weight ration at elevation is 2:1
- At sea level -> 2.7:1

- From lift equation, we took V_{3000} to be 17% greater than at sea level

AVIONICS SUITE

- Flight controller- Pixhawk 6c
- Firmware- Ardupilot
- Navigation – Here4 GNSS RTK module
- LiDAR - Lightware SF11/C
- Payload Release Mechanism- High Torque DC Planetary geared motor, Braided dyneema cable, Machined aluminium pulley, S beam load cell sensor, Electronically controlled mechanical servo release latch
- Telemetry- Holybro Telem long rang radio (915MHz)

POWER

Li-Po soft case battery pack in 12S1P (44.4V)

Total capacity = 25000 mAh

WEIGHT DISTRIBUTION

- Payload = 10 kg
- Battery pack = 3.3 kg
- Airframe and structures (mostly carbon fiber/composite) = 6 kg
- Rotors = 4 kg
- Avionics suite = 1.5 kg

Total is below 25kg limit

AERODYNAMICS AND STRUCTURAL REASONING

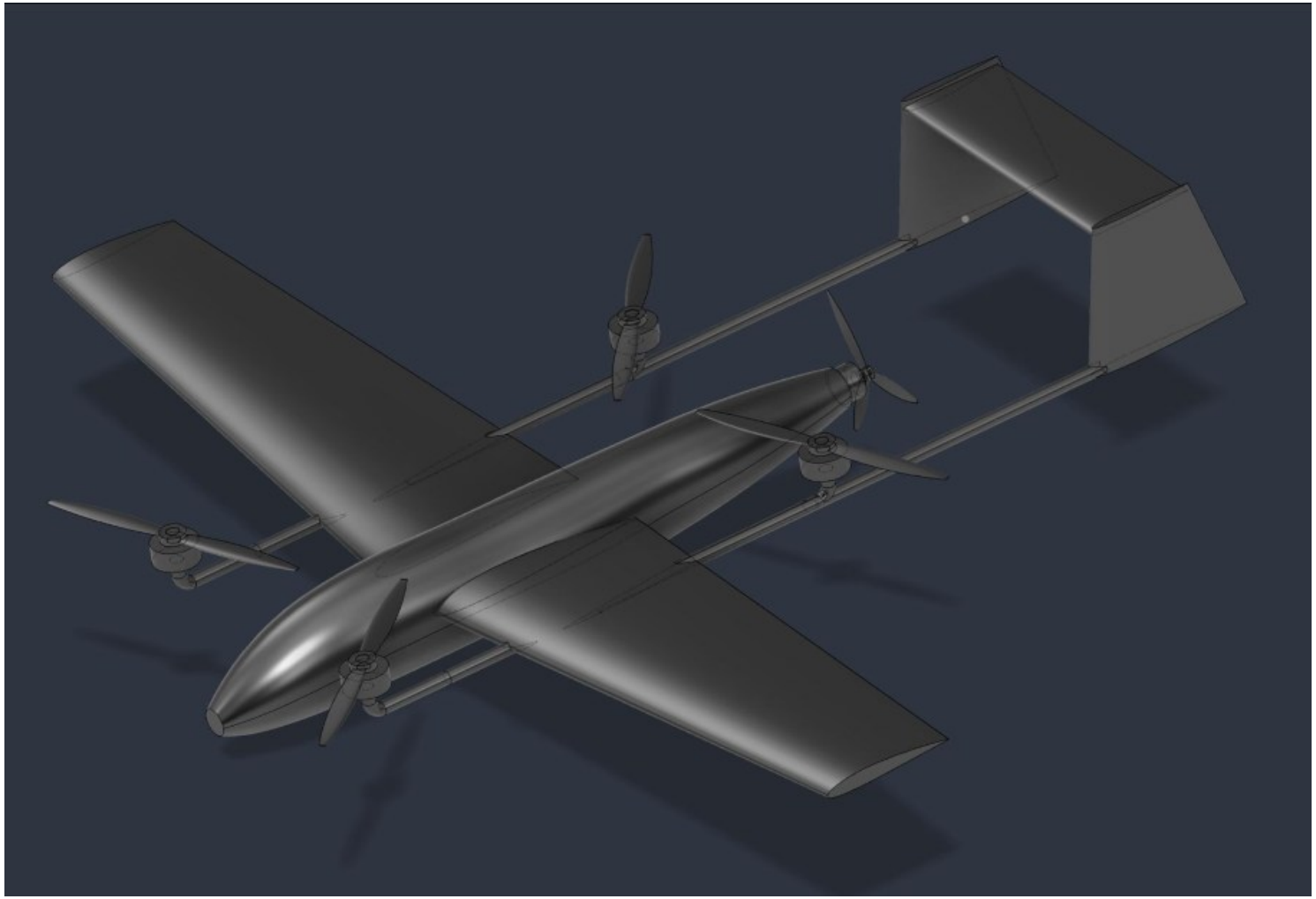
Wingspan and Chord length assumptions :-

- Aspect Ratio of wing = 7
- Taper ratio of root to chord length = 0.6
- Wing Loading = 20kg/m^2
- Zero Swipe wing

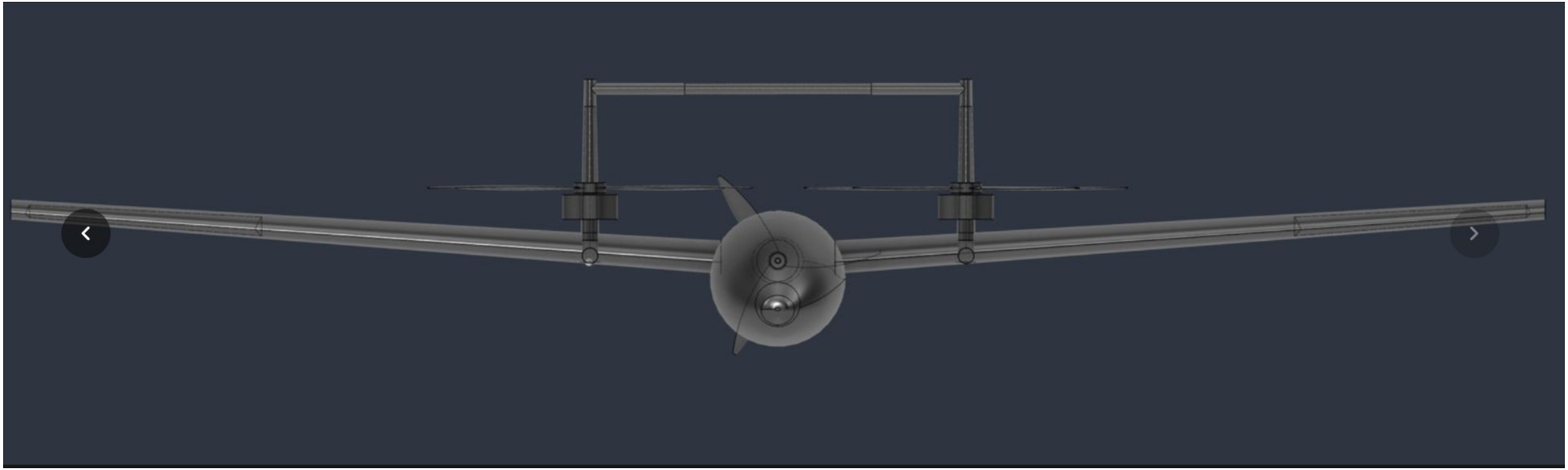
Tail Sizing :-

- Horizontal tail volume ratio =0.5
- CG location at 0.3 of chord length from the leading edge.
- Tail efficiency factor of 0.90.
- Vertical tail volume of 0.04.
- The vertical tail plane is perpendicular to our fuselage axis.
- The vertical tail taper ratio of 0.6.
- A vertical tail arm of 1.5

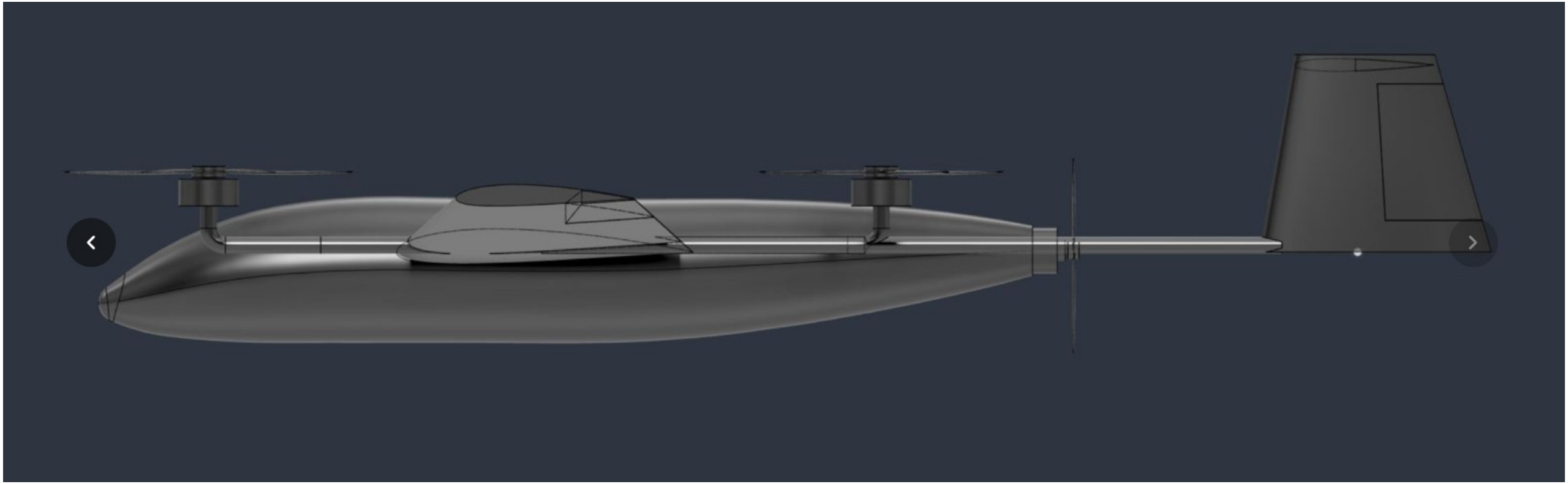
CAD DESIGN



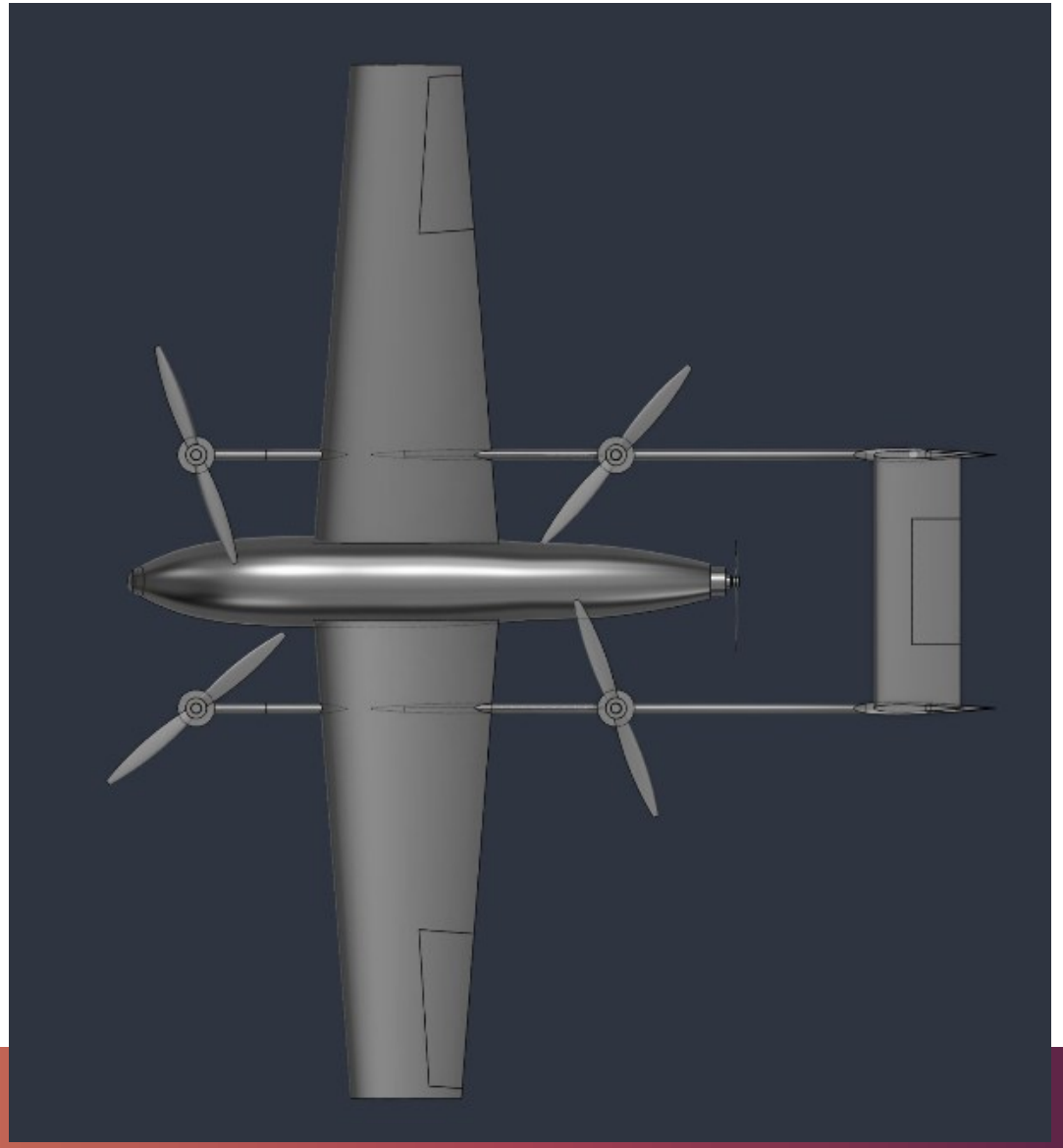
CAD DESIGN



CAD DESIGN



CAD DESIGN




THANK YOU